Exact degrees for even Catalan-power Hankel determinants

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Abstract

For integers $j \geq 1$ and $k \geq 2j-1$, the signed Hankel determinant sequences formed from the coefficients of the even Catalan power $C(z)^{2k}$, with zero moment shift and matrix sizes $kn+j$ and $kn+k+1-j$, agree for every $n \geq 0$ with rational polynomials of exact degree $(2j-1)(k-j)$. The sign is $(-1)^{n\binom{k}{2}}$. A quotient-minor formula gives the degree bound; a strictly positive moment determinant proves that its leading term does not cancel. We give the full proof, including the boundary cases, and exact finite verification code. The result concerns these two offsets in the stated parameter range, not arbitrary moment shifts, odd Catalan powers, or Narayana-parameter deformations.

Statement and notation

The residue-three case

Let

$$C(z) = \frac{1-\sqrt{1-4z}}{2z}, \qquad c_m^{(2k)} = [z^m]C(z)^{2k} = \frac{2k}{2m+2k}\binom{2m+2k}{m},$$

and

$$D_{2k,0}(N) = \det(c_{u+v}^{(2k)})_{0\leq u,v<N}, \qquad D_{2k,0}(0) = 1.$$

For integers $k \geq 5$, $n \geq 0$, and $r \in \{3, k-2\}$, set

$$f_{k,r}(n) = (-1)^{nK} D_{2k,0}(kn+r), \qquad K = \binom{k}{2}.$$

The target is the existence of polynomials $F_{k,r} \in \mathbb{Q}[n]$ such that

$$F_{k,r}(n) = f_{k,r}(n) \quad (n \geq 0), \qquad \deg F_{k,r} = 5(k-3).$$

At $k = 5$, the two residues are identical. The asserted degree is exact, not merely an upper bound. The polynomials are integer-valued on the indicated integer arguments; integer coefficients are not asserted.

This is precisely the $j = 3$ specialization of [C23, Conjecture 6, equation (19)], with the sign convention in equation (16). The specializations with $5 \leq k \leq 15$ are already included in [WX19, Theorem 11].

Theorems and proofs

1. Resolution and leading coefficient

Theorem 1. The target holds for every $k \geq 5$ and every $n \geq 0$. Define

$$b_m = \frac{2\zeta(2m)}{\pi^{2m}} = \frac{(-1)^{m+1}2^{2m}B_{2m}}{(2m)!} > 0 \qquad (m \geq 1), \tag{1}$$

where $u/(e^u - 1) = \sum_{m\geq 0} B_m u^m/m!$. The leading coefficients are

$$\boxed{[n^{5(k-3)}]F_{k,3}(n) = L_k := k^{4(k-3)}\big(b_{k-4}b_{k-2} - b_{k-3}^2\big) > 0,} \tag{2}$$

$$\boxed{[n^{5(k-3)}]F_{k,k-2}(n) = (-1)^K L_k.} \tag{3}$$

In particular, there is no unproved noncancellation assumption in the degree claim. Strict positivity in (2) is established below.

The proof also extends to every residue pair in the stated range of [C23, Conjecture 6]. Section 7 gives a complete proof of that stronger conclusion, not just a proposed extension. No claim of publication priority is made.

2. Moment functional and a finite determinant reduction

Put $s = k - 1$. Define monic polynomials by

$$P_0(x) = 1, \quad P_1(x) = x - 2, \quad P_{m+1}(x) = (x-2)P_m(x) - P_{m-1}(x). \tag{4}$$

Lemma 2.1 (moment bridge). There is a linear functional $\mathcal{L}$ on $\mathbb{Q}[x]$ such that

$$\mathcal{L}(P_aP_b) = \delta_{ab}, \qquad c_m^{(2k)} = \mathcal{L}\big(x^m Q(x)\big), \qquad Q(x) = x^s P_s(x). \tag{5}$$

Moreover, $H_N(\mathcal{L}) := \det(\mathcal{L}(x^{u+v}))_{u,v<N} = 1$.

Proof. Make the P_m an orthonormal basis of the polynomial space. The recurrence (4) says that multiplication by x is symmetric: its matrix has diagonal entries 2 and adjacent off-diagonal entries 1. At the boundary, $xP_0 = P_1 + 2P_0$. Define $\mathcal{L}(h) = \langle 1, h\rangle$. Symmetry of multiplication gives $\langle f, g\rangle = \mathcal{L}(fg)$, proving orthogonality.

The coefficient of P_a in x^m counts nonnegative Motzkin walks of length m from height 0 to height a, with horizontal-step weight 2 and both sloping-step weights 1. Let $G(z)$ be the excursion generating function. Decomposition at the first step and first return gives

$$G = 1 + 2zG + z^2G^2.$$

This has a unique formal solution with constant coefficient 1. Since $C = 1 + zC^2$, its solution is $G = C^2$.

Decomposing a walk ending at height s at its last upward departures from heights $0, \ldots, s-1$ gives s up-steps and $s+1$ excursions. Thus

$$\sum_{m\geq 0} \mathcal{L}(x^m P_s)z^m = z^s G(z)^{s+1}.$$

Taking the coefficient of z^{m+s} proves (5), because $s+1=k$. Finally, the change of basis from monomials to the monic P_m is unit triangular. Their Gram matrix is the identity, so $H_N(\mathcal{L})=1$, including the empty determinant. $\square$

For any integer $N \geq 0$, let $\mathcal{R}(N)$ be the determinant of the coefficient vectors of

$$P_N, P_{N+1}, \ldots, P_{N+2s-1} \pmod{Q}$$

in the ordered monomial basis $1, x, \ldots, x^{2s-1}$.

Lemma 2.2 (remainder determinant). For every $N \geq 0$,

$$D_{2k,0}(N) = \mathcal{R}(N). \tag{6}$$

Proof. Consider the $(N+2s)$-column coefficient matrix of

$$QP_0, \ldots, QP_{N-1}, P_N, \ldots, P_{N+2s-1}.$$

In the orthonormal P-basis, its last $2s$ columns are the corresponding unit vectors. Its determinant is therefore $\det(\mathcal{L}(P_uQP_v))_{u,v<N} = H_N(Q\mathcal{L})$. Changing the first N monic basis polynomials to monomials does not change this determinant. By Lemma 2.1 it is $D_{2k,0}(N)$.

For each last column, division by Q has quotient degree at most $N-1$. Subtract a linear combination of the first N columns to leave its remainder. Move these $2s$ remainder columns to the front. The sign is $(-1)^{2sN}=1$. The resulting matrix has remainder block in its upper left, zero lower-left block, and a unit-triangular lower-right block. Its determinant is $\mathcal{R}(N)$. No nonvanishing of $H_N(Q\mathcal{L})$ was assumed. $\square$

This reduction is a specialization of the classical orthogonal-polynomial moment-modification identity in [K21, Theorem 1 and Proposition 5]. The proof above is included to check its hypotheses and conventions directly; the reduction itself is not asserted to be new.

3. Evaluation matrix, polynomiality, and reflection

The roots of P_s are

$$\alpha_i = 2 + 2\cos(i\pi/k), \qquad 1 \leq i \leq s.$$

They are distinct and strictly positive. In particular they are different from zero. Let

$$V(\alpha) = \prod_{1 \leq i < j \leq s} (\alpha_j - \alpha_i).$$

For a polynomial of degree below $2s$, take its values at the s roots, followed by its coefficients of degrees $0, \ldots, s-1$. The determinant of this linear transformation on the monomial basis is

$$V(\alpha)P_s(0)^s \neq 0. \tag{7}$$

Indeed, move the s coefficient rows to the front. Their block is $(I_s\ 0)$; the remaining block in degrees $s, \ldots, 2s-1$ has determinant $(\prod_i \alpha_i^s)V(\alpha)$. The row-move sign $(-1)^{s^2}$ gives exactly (7), since $P_s(0) = (-1)^s \prod_i \alpha_i$.

A multiple of $Q = x^sP_s$ is annihilated by all these functionals. Consequently, if E_N is the matrix with columns $P_N, \ldots, P_{N+2s-1}$, root-value rows first and coefficient rows second, then

$$\mathcal{R}(N) = \frac{\det E_N}{V(\alpha)P_s(0)^s}. \tag{8}$$

Extend (4) to all integer indices. It gives

$$P_{-1} = 0, \qquad P_{-m-2} = -P_m \quad (m \geq 0). \tag{9}$$

Use the remainder determinant to define $\mathcal{R}(N)$ also for negative N. This is an algebraic extension, not a negative-size Hankel determinant. Two useful identities, valid for integer m, are

$$P_m(\alpha_i) = \frac{\sin((m+1)i\pi/k)}{\sin(i\pi/k)}, \qquad [x^a]P_m(x) = (-1)^{m-a}\binom{m+a+1}{2a+1}. \tag{10}$$

The binomial in the second identity is the falling-factorial polynomial. For nonnegative m, the identities follow from (4) and its initial conditions. For negative m, they follow from (9) and $\binom{-u}{v} = (-1)^v\binom{u+v-1}{v}$; the index -1 gives zero.

Lemma 3.1. For any fixed integer residue r,

$$(-1)^{nK}\mathcal{R}(kn+r) \quad (n \in \mathbb{Z})$$

is the restriction of a polynomial $F_{k,r} \in \mathbb{Q}[n]$ of degree at most s^2.

Proof. By (10), root row i of E_{kn+r} has a common factor $(-1)^{ni}$. Their product is $(-1)^{nK}$. Removing it makes the root rows constant in n.

Coefficient row a, $0 \leq a < s$, has a common sign $(-1)^{kn}$; after removing it, each entry is a polynomial of degree at most $2a+1$. The product of these signs is 1, because $kns = k(k-1)n$ is even. The determinant in (8) thus has degree at most $\sum_{a=0}^{s-1}(2a+1) = s^2$. It agrees with the signed remainder determinant at every integer n, including negative integers. Its coefficients are rational: its values at any s^2+1 distinct integers are integers, and the corresponding Vandermonde interpolation system is over $\mathbb{Q}$. □

Lemma 3.2 (reflection). As polynomial identities,

$$F_{k,k+1-r}(n) = (-1)^{s+K}F_{k,r}(-n-3). \tag{11}$$

Proof. Reverse the $2s$ columns and apply (9). The $2s$ individual minus signs cancel, and reversal has sign $(-1)^s$. Therefore

$$\mathcal{R}(-N-2s-1) = (-1)^s\mathcal{R}(N).$$

If $N = kn+r$, its reflected index is

$$-N-2s-1 = k(-n-3) + (k+1-r).$$

Combining the two signing factors gives $(-1)^{s+K}$. Lemma 3.1 promotes the identity on integer arguments to the polynomial identity (11). □

4. The residue-3 quotient identity

For $n \geq 0$, write

$$N = kn+3, \qquad M = k(n+1), \qquad p(x) = P_{M-1}(x), \quad q(x) = P_{M+k-1}(x),$$

$$\frac{q(x)}{p(x)} = \sum_{a \geq 0} h_a(k, M)x^a. \tag{12}$$

The quotient is a well-defined formal series because $p(0) = (-1)^{M-1}M \neq 0$.

Lemma 4.1. For every $k \geq 5$ and $n \geq 0$,

$$\boxed{F_{k,3}(n) = -(n+1)^s \left(h_{s-2}(k,M)^2 - h_{s-3}(k,M)h_{s-1}(k,M)\right).} \tag{13}$$

Proof. Let $\ell = M - 1 = kn + s$. In the column window $P_N, \ldots, P_{N+2s-1}$, replace the columns with leading indices $\ell, \ldots, \ell + s - 3$ by

$$p, xp, \ldots, x^{s-3}p.$$

These are determinant-preserving triangular column operations. Specifically, $x^a P_\ell = P_{\ell+a} +$ a combination of $P_{\ell-a}, \ldots, P_{\ell+a-1}$, and $\ell - a \geq N$ for $a \leq s - 3$.

The final two original columns are $q = P_{kn+2s+1}$ and $P_{kn+2s+2}$. Replace the latter by

$$xq = P_{kn+2s+2} + 2P_{kn+2s+1} + P_{kn+2s}.$$

This is again determinant-preserving. Both p and q vanish at every α_i, because their indices plus one are multiples of k.

The s columns which remain nonzero in root rows are

$$P_{kn+3}, \ldots, P_{kn+s-1}, \quad P_{kn+2s-2}, P_{kn+2s-1}, P_{kn+2s}.$$

After removing row factors $(-1)^{ni}$, these become

$$P_3, \ldots, P_{s-1}, -P_2, -P_1, -P_0.$$

Their root determinant is $(-1)^{s-3}V(\alpha)$. To see the sign explicitly: there are three minus signs, $3(s-3)$ crossings to move the last three polynomials to the front, and three crossings to reverse their order.

Moving the three right-hand root columns across the $s - 2$ columns made from p has sign $(-1)^{3(s-2)}$. Multiplying these two signs gives $(-1)^{s-3+3(s-2)} = -1$. After all root columns have been put first, the lower-right coefficient block has columns

$$p, xp, \ldots, x^{s-3}p, \;\; q, xq.$$

Multiply the truncated coefficient vectors by $1/p$, modulo x^s. The determinant of this multiplication is $p(0)^{-s}$, since its matrix is triangular. The first $s - 2$ columns become the corresponding coordinate vectors, leaving the minor

$$\begin{vmatrix} h_{s-2} & h_{s-3} \\ h_{s-1} & h_{s-2} \end{vmatrix}.$$

The denominator in (8) supplies $P_s(0)^s$, and

$$\left(\frac{p(0)}{P_s(0)}\right)^s = \left((-1)^{kn}(n+1)\right)^s = (n+1)^s.$$

The root-row sign is exactly the one already removed in the definition of $F_{k,3}$. This proves (13), including $n = 0$. □

5. Exact degree and strict noncancellation

The classical partial fraction expansion of cotangent [DLMF, 4.22.3] is

$$\cot u = \frac{1}{u} + 2u \sum_{\nu \geq 1} \frac{1}{u^2 - \pi^2 \nu^2}.$$

Expanding for $0 < |u| < \pi$ gives

$$\cot u = \frac{1}{u} - \sum_{a \geq 1} b_a u^{2a-1}, \qquad b_a = 2 \sum_{\nu \geq 1} (\pi \nu)^{-2a}. \tag{14}$$

The expansion and interchange of sums are justified by absolute uniform convergence on compact subdisks of radius below π, after removal of the simple pole. The Bernoulli expression (1) is the equivalent Laurent coefficient formula [DLMF, 4.19.6].

Lemma 5.1. For fixed k and every $a \geq 1$, $h_a(k, M)$ is a Laurent polynomial in M with no exponent below -1, with degree $2a - 1$, and

$$[M^{2a-1}] h_a(k, M) = (-1)^{k+1} k b_a. \tag{15}$$

Proof. Make the formal substitution

$$x = 2 - 2\cos\theta = \theta^2 + O(\theta^4).$$

More precisely, θ^2 is the unique series in $x\mathbb{Q}[[x]]$ inverting this substitution, and has leading coefficient 1. From (4),

$$P_m(2 - 2\cos\theta) = (-1)^m \frac{\sin((m+1)\theta)}{\sin\theta}.$$

Consequently (12) gives

$$\frac{q}{p} = (-1)^k \frac{\sin((M+k)\theta)}{\sin(M\theta)} = (-1)^k \left(\cos(k\theta) + \sin(k\theta)\cot(M\theta)\right). \tag{16}$$

This identity holds for the positive integer values of M used in the Hankel determinant, and its right side defines the required coefficient extension in the formal ring $\mathbb{Q}[M, M^{-1}][[x]]$.

In (14), the pole contributes only a term proportional to M^{-1}. The cosine contributes a term independent of M. For the coefficient of x^a, only cotangent terms indexed by $1 \leq v \leq a$ can contribute, because $\sin(k\theta)\theta^{2v-1}$ starts with $k\theta^{2v}$. Thus the possible exponents of M are $-1, 0, 1, 3, \ldots, 2a-1$. The highest one comes only from $v = a$, and

$$[x^a]\left(\sin(k\theta)\theta^{2a-1}\right) = k.$$

This proves (15), with a nonzero coefficient. □

Now put $t = k - 3 = s - 2$. The hypothesis $k \geq 5$ is exactly what ensures $t - 1 \geq 1$, so Lemma 5.1 applies to all three coefficients in (13). Their products have no pole worse than M^{-2}, and have degree at most $4t - 2$. Multiplication by $(n+1)^s = (M/k)^s$ clears all possible poles. The degree is at most

$$s + 4t - 2 = 5(k - 3).$$

By (15), the coefficient at that degree is

$$-k^{4t}\left(b_t^2 - b_{t-1} b_{t+1}\right) = L_k.$$

It remains to prove that this coefficient does not vanish. Put $u_\nu = (\pi\nu)^{-2}$. Because $t \geq 2$, all sums below converge absolutely, and (14) gives the exact identity

$$b_{t-1}b_{t+1} - b_t^2 = 4 \sum_{1 \leq a < b} u_a^{t-1} u_b^{t-1} (u_a - u_b)^2 > 0. \tag{17}$$

Every term is nonnegative, and already the term $(a, b) = (1, 2)$ is positive. This proves (2) and proves that the degree is exact. No finite computation is needed for this strict inequality.

Finally, apply (11) with $r = 3$. Reflection preserves degree, and the leading coefficient changes by

$$(-1)^{s+K+5(k-3)} = (-1)^K,$$

because $s + 5(k - 3) = 6k - 16$ is even. This proves (3) and completes Theorem 1. □

6. Boundary tests and exact verification

At the smallest permitted parameter $k = 5$, the two residues coincide. The quotient formula gives the following exact polynomial:

$$\begin{aligned} F_{5,3}(n) = \frac{125}{3024}(n+1)^2(n+2)^2(2n+3)^2 \\ \cdot (500n^4 + 3000n^3 + 5125n^2 + 1875n - 1092). \end{aligned} \tag{18}$$

It has degree 10, leading coefficient $15625/189$, and value -1625 at $n = 0$, equal to the original 3×3 determinant. Its reflection symmetry is $F_{5,3}(n) = F_{5,3}(-n - 3)$, as required.

The lower bound on k must not be dropped. The excluded pair $k = 4, j = 3$ has degree 6, not $5(k - 3) = 5$; it is the reflected residue of $j = 2$. This agrees with the printed small-parameter formulas in [WX19, Theorem 6] and [C18, page 30]. It is not a counterexample to Theorem 1, whose range is $k \geq 5$.

The accompanying standard-library program `verify_cigler.py` performs exact integer, rational, and Laurent-polynomial arithmetic. It does not use floating point, a numerical zeta evaluator, or polynomial fitting. Its deterministic data files contain the following recorded checks. The complete suite was not rerun when preparing this research package; the counts below describe the included records and reproducible code.

Check	Recorded range / count
Symbolic residue-3 polynomials	23 values of k: 5 through 24, and $32, 40, 50$
Original Hankel determinant comparisons	60, for $5 \leq k \leq 12$, $0 \leq n \leq 3$, both distinct target residues
Independent remainder-determinant comparisons	429, for $5 \leq k \leq 24$, $-4 \leq n \leq 6$, both distinct target residues
General-j quotient comparisons	267, for $2 \leq k \leq 18$, every $1 \leq j \leq \lfloor (k+1)/2 \rfloor$, $0 \leq n \leq 2$
Supplementary $k = 16$ certificate	All 452 values and both Newton coefficient lists reproduced
Empty Hankel determinant	Checked to equal 1

The negative-index checks use the algebraic remainder determinant, never a negative-size Hankel determinant. They include $n = -1$, where the raw quotient has a zero denominator but its polynomial

extension is well-defined. The symbolic check constructs the coefficients of $\binom{M+a}{2a+1}$ as polynomial products and carries out formal series division, independently of the trigonometric derivation of their top degree.

7. Stronger theorem: the full stated residue-degree range

The same construction proves the following generalization of the $j = 3$ case.

Theorem 7.1. Let $j \geq 1$ and $k \geq 2j - 1$ be integers. For every $n \geq 0$, the two signed sequences in residues j and $k + 1 - j$ agree with polynomials of exact degree

$$(2j - 1)(k - j). \tag{19}$$

For $k \geq 2$, set

$$d = j - 1, \qquad t = k - j, \qquad m_0 = k - 2j + 2.$$

If $d \geq 1$, the leading coefficient of $F_{k,j}$ is

$$\boxed{(-1)^{kd} k^{2dt} \det\big(b_{m_0+a+b}\big)_{0\leq a,b<d}.} \tag{20}$$

The determinant in (20) is strictly positive. For $d = 0$, the determinant is interpreted as 1, and the leading coefficient is 1.

Proof. The case $k = j = 1$ is immediate from the excursion moments in Lemma 2.1: $D_{2,0}(N) = 1$. Suppose henceforth that $k \geq 2$.

Take $N = kn + j$, $\ell = kn + s = M - 1$, and again use $p = P_\ell$ and $q = P_{kn+2s+1}$. Replace the columns starting at P_ℓ by

$$p, xp, \ldots, x^{s-j}p$$

and the columns starting at q by

$$q, xq, \ldots, x^{j-2}q.$$

For $j = 1$, the second list is empty. Each replacement is a unit-triangular column operation: the smallest required index in the first list is $\ell-(s-j) = N$; in the second list it is $kn+2s+1-(j-2) = kn+2s-j+3$, which is in the remaining right-hand window. The upper bounds on required indices are exactly the columns being replaced. Thus no polynomial outside the original window is introduced.

After removal of root-row signs, the surviving s root columns are

$$P_j, \ldots, P_{s-1}, -P_{j-1}, \ldots, -P_0.$$

Their determinant relative to $V(\alpha)$ has sign

$$(-1)^{j+j(s-j)+\binom{j}{2}}.$$

Moving the rightmost j root columns across the $s - j + 1 = s - d$ columns made from p has sign $(-1)^{j(s-j+1)}$. The combined sign is therefore $(-1)^{\binom{j}{2}}$.

Divide the lower coefficient block by p, modulo x^s, just as in Lemma 4.1. Its first $s - d$ columns become coordinate vectors, while its last d columns become $h, xh, \ldots, x^{d-1}h$, where $h = q/p$. Thus the exact quotient identity is

$$F_{k,j}(n) = (-1)^{\binom{j}{2}}(n + 1)^s \det\big(h_{t+a-b}(k, M)\big)_{0\leq a,b<d}. \tag{21}$$

For $d = 0$, it gives $F_{k,1}(n) = (n+1)^s$, with the asserted exact degree.

For $d \geq 1$, the least coefficient index in (21) is

$$t - (d-1) = m_0 = k - 2j + 2 \geq 1.$$

Therefore Lemma 5.1 applies to every entry. Each determinant term has a possible pole of order at most d, cleared by M^s because $s \geq d$. Its degree in M is at most

$$\sum_{a=0}^{d-1} \bigl(2(t + a - \sigma(a)) - 1\bigr) = d(2t-1)$$

for every permutation σ. Since $s = t + d$, the total degree is at most

$$s + d(2t-1) = (2d+1)t = (2j-1)(k-j).$$

The coefficient at this degree in the inner determinant is

$$(-1)^{d(k+1)} k^d \det(b_{t+a-b})_{0 \leq a,b < d}.$$

Reversing its columns gives

$$\det(b_{t+a-b}) = (-1)^{\binom{d}{2}} \det(b_{m_0+a+b}).$$

Combining this sign with $(-1)^{\binom{d+1}{2}}$ from (21), and replacing M by $k(n+1)$, gives exactly (20). The sign simplification is

$$\binom{d+1}{2} + d(k+1) + \binom{d}{2} \equiv kd \pmod 2.$$

To prove strict positivity, let $v = (v_0, \ldots, v_{d-1}) \in \mathbb{R}^d$ be nonzero. Formula (14) yields

$$\sum_{a,b=0}^{d-1} v_a v_b b_{m_0+a+b} = 2 \sum_{\nu \geq 1} u_\nu^{m_0} \left(\sum_{a=0}^{d-1} v_a u_\nu^a \right)^2 > 0. \tag{22}$$

The series converges since $m_0 \geq 1$ and the polynomial in parentheses is bounded on the bounded set of points u_ν. A nonzero polynomial has only finitely many real zeros, whereas the u_ν are infinitely many distinct positive numbers. Hence at least one summand is positive. This proves that the matrix in (20) is positive definite, so its determinant is strictly positive. The degree bound is consequently attained.

Finally, (11) transfers the same exact degree to residue $k + 1 - j$. This proves (19) throughout its stated range. □

8. Hypotheses, scope, and attribution

Determinant convention. The coefficient sequence is C^{2k}, the shift is zero, and the Hankel size is exactly $kn + r$. The sign is $(-1)^{n\binom{k}{2}}$. Theorem 1 treats both specified residues and all $n \geq 0$, with no exceptional $k \geq 5$. The case $k = 5$ is included once, not assumed to consist of two different residues.

Theorem hypotheses. The base polynomials are monic and orthonormal for the explicitly constructed functional, whose norms are all 1. The modifying polynomial is monic of even degree. Its simple nonzero roots are distinct from the confluent point 0, so (7) is nonzero. Series division uses $M = k(n+1) \neq 0$ in the target range. The only analytic input is the classical cotangent expansion; it is used inside its

disk of convergence to identify coefficients, and then as a formal coefficient identity. Every sum in the strict-positivity argument converges.

Exactness. The positive coefficient in (17), rather than finite interpolation or a proposed degree bound, establishes exact degree. The algebraic negative-index extension is used solely for the proved polynomial reflection identity. No negative-size Hankel determinant is used.

Scope. Theorem 1 covers the stated residue-three case. Theorem 7.1 further proves the full stated j-range in [C23, equation (19)]. The argument does not assert formulas for arbitrary shifts, odd Catalan powers, or Narayana-parameter deformations.

Method attribution. The moment-functional construction, determinant reduction, polynomiality, and reflection form the preliminary framework and are included with proofs. The classical moment-modification identity is attributed to [K21] in Section 2. The higher-rank quotient minor and the strictly positive moment determinant supply the uniform exact-degree argument. The supplementary finite certificate is not a premise of this proof.

Literature scope. [C23] and [WX19] identify the target and known finite range. [C18] supplies the neighboring Narayana statements. [K21] gives the classical determinant transformation. [H26] treats shifts $-m, 1-m, 2-m, 3-m$, which do not give the present unshifted target for $m = k \geq 5$. The additional 2026 source [CS26, Theorem 1.4], specialized to its parameters $k_{\mathrm{CS}} = k-1, m = k-1, c = 2$, gives the neighboring residue-0 and residue-1 formulas; it does not state the residue-3 degree theorem. No pre-existing uniform proof of the exact residue-degree statement was identified in these checked sources. This is a limited literature check, not a worldwide priority claim.

AI assistance. This work was prepared with AI assistance. No independent human peer-review or proof-assistant certification is claimed.

References

[**C23**] Johann Cigler, *Some experimental observations about Hankel determinants of convolution powers of Catalan numbers*, arXiv:2308.07642v2 (2023). Equations (1)–(2), page 1; equation (16) and Conjecture 6, equation (19), page 4. This is the exact source of the target and its range.

[**WX19**] Ying Wang and Guoce Xin, *Hankel determinants for convolution powers of Catalan numbers*, arXiv:1811.00248v2; *Discrete Mathematics* 342 (2019), 2694–2716. Conjecture 10, page 3, and Theorem 11, page 4. Only the specializations in the range specified by [C23] are used here.

[**C18**] Johann Cigler, *Catalan numbers, Hankel determinants and Fibonacci polynomials*, arXiv:1801.05608v3 (2018). Conjecture 7.7, equation (7.13), page 30. This concerns a neighboring problem, not the residue-three statement.

[**K21**] Christian Krattenthaler, *Hankel determinants of linear combinations of moments of orthogonal polynomials, II*, arXiv:2101.04225v5 (2021). Theorem 1 and Proposition 5. DOI: 10.1007/s11139-021-00514-8.

[**H26**] Guo-Niu Han, *Hankel Determinants for Convolution Powers of Narayana Polynomials*, arXiv:2512.12603v1; *Electronic Journal of Combinatorics* 33(2) (2026), P2.42. Theorems 4–7. DOI: 10.37236/14995.

[**CS26**] Shane Chern and Wenle Shi, *Hankel determinants of Catalan-like sequences*, arXiv:2608.27208v1 (2026). Theorem 1.4, page 3.

[**DLMF**] NIST Digital Library of Mathematical Functions, equations 4.22.3 (cotangent partial fractions) and 4.19.6 (cotangent Laurent coefficients).